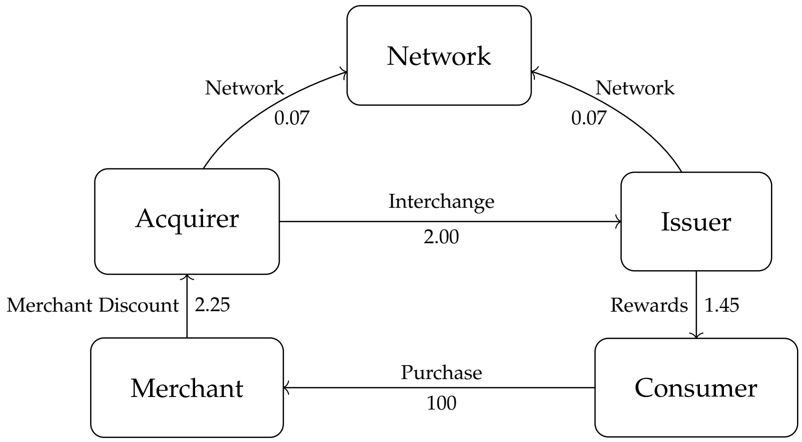
Figure 1: Payment flows in a payment network.



Notes: Representative credit card fees in 2019. The merchant discount fee comes from Nilson (2020). The average network fee comes from example rate sheets from acquirers and from dividing the non-foreign exchange fees from Visa's 10k by the total payment volumes (Visa 2020; Helcim 2021). I split the network fees evenly between the two sides as in (Federal Reserve 2010). The interchange is derived from Visa's interchange schedule as the average of the rates for Visa Signature and Visa Infinite cards at a large retailer (Visa 2019). The rewards are from large banks' annual reports in 2019.

Payment networks are two-sided platforms that simultaneously set prices for merchants and consumers. While AmEx sets merchant fees and consumer rewards directly, "open-loop" networks like Visa and MC do so indirectly, through the *interchange fee* and *network fee*.

Visa and MC connect four parties: merchants, acquirers (merchants' banks), issuers (consumers' banks), and consumers (Benson et al. 2017). Figure 1 shows the payment flows with representative prices. When a consumer uses her credit card to buy \$100 of product at a large retailer, the merchant pays a \$2.25 merchant discount fee to her acquiring bank to process the transaction. The acquirer can be a bank like Wells Fargo or a fintech firm like Square. Of the merchant discount, around \$2 goes to the issuing bank (e.g., Chase) as interchange. The issuer and the acquirer collectively pay around \$0.14 in network fees to Visa. Financial reports show that cardholder rewards average around \$1.45 per \$100 of transaction value. [1]

Regulatory shocks confirm that interchange strongly affects merchant fees and rewards, but not borrowing costs. When the E.U. and Australia reduced interchange by regulation, merchant fees declined roughly one-for-one (Gans 2007; Valverde et al. 2016; European Commission 2020). Appendix Figure **??** shows that after Australia capped credit card interchange, rewards fell and consumer annual fees on rewards cards rose, while annual fees on non-reward cards and interest rates were unchanged. Interchange has little effect on interest rates because balance-carriers pay 70% of interest but generate

[1] These figures are similar to the regulatory numbers reported in Drechsler et al. (2025), which finds that banks' interchange income averages 1.82% of purchase volume while rewards costs average 1.57%.

only 10% of purchase volume (Adams et al. 2022).